
Structural Predictors of Chronic Absenteeism in NYC Schools

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Abstract

Chronic absenteeism may pose as a significant issue for students, and its leading structural predictors remain difficult to isolate. This study frames the prediction of school-level chronic absenteeism percentages in NYC students as a supervised regression task, using demographic and socioeconomic features. We first merged existing attendance and demographic datasets into one that spanned the years 2013 to 2018. It was partitioned using a temporal split as it was more realistic for this scenario. 2013-2016 for training, 2016-2017 for tuning, and 2017-2018 for testing. Predictive performance was first evaluated across two baseline models, those being Multiple Linear Regression (MLR) and polynomial regression on the Economic Need Index (ENI), as well as two machine learning algorithms: Random Forest (RF), and eXtreme Gradient Boosting (XGBoost). Baseline analysis revealed a minimized tune error at degree 9 despite degrees 3-4 having smaller gaps between splits, and also data limitations, including 1,493 missing ENI values and a systemic 2017 shift in poverty reporting metrics (causing a increase in values in our ENI and % poverty). The missing ENI values were not as significant due to our train split years and the three other predictors; the change in ENI reporting had more weight on the model.

The RF mode identified the most significant predictor as students with disabilities. However, comparing it to XG Boost where the highest predictor was ENI, we found RF has a tendency to memorize training distributions rather than generalize well to all data.

as a supervised regression task. Given school-level demographic and socioeconomic features from publicly available data from previous years, we aim to predict the percentage of students who are chronically absent.

The objective of this project is to uncover the leading structural predictors of chronic absenteeism in NYC public schools. Studying this data provides critical context for how chronic absenteeism affects student outcomes. However, the focus of our model is strictly identifying the most significant drivers of absenteeism itself.

Since the proposal, we have completed data ingestion, cleaning, and merging of the attendance and demographic datasets from 2013 to 2018. To reflect a realistic forecasting scenario, we implemented a temporal split (2013–2016 training; 2016–2017 developmental; 2017–2018 testing). Our approach began with two baseline models, multiple linear regression (MLR) and polynomial regression on the Economic Need Index (ENI) up to degree 10. It evolved eventually into more complex machine learning models including Random Forest and XGBoost. We chose to implement these models as they provided superior predictive accuracy and helped to identify the most influential predictors on chronic absenteeism.

Multiple key data limitations were identified: ENI as a metric was introduced in the 2014–2015 school year and substantially affected by a methodological change in the 2017–2018 school year. This resulted in 1,493 missing values for the 2013–2014 school year, and a large discrepancy between tune (2016–2017) and test (2017–2018) error. Although the missing ENI values were mitigable due to our train split containing ENI for 2014–2015 and 2015–2016, and having three other predictors, the issue of the ENI calculation itself changing is unresolvable, and its implications are explored further in the diagnostics section.

1. Introduction

Chronic absenteeism remains a significant barrier to student success, with structural drivers, including socioeconomic disadvantage and demographic inequality, persisting beyond pandemic-related disruptions. New York City, with approximately 900,000 publicly enrolled students, provides an ideal setting to study these factors at scale. We framed this project

2. Related Work

In a study titled “Predicting and Analyzing Student Absenteeism Using Machine Learning Algorithm,” researchers Lindita Mukli and Amarildo Rista utilized a machine learning algorithm to analyze attributes of university students’ lives and its potential correlation to attendance. They compiled a dataset of 210,000 records of the teaching hours of 500 students, grouped by 5 categories and 26 attributes, during an academic year. Research concluded that certain attributes were more significant contributors to increased absences than others, usually those involving personal matters. Other influential attributes included employment while enrolled and longer commutes, among others (Mukli & Rista, 2022). A key limitation here is that these results are based on a dataset with only one university. Thus, this study cannot be used to make generalizations about drivers of absenteeism in other educational institutions.

In another report titled “NY Attendance Trends 18-19”, Leslie Vald aims to identify how to minimize low attendance days (“dip” days) by analyzing a Kaggle dataset on daily attendance of New York students. Vald uses Big Query to clean the data’s start and end dates and further analyzes absences in relation to time, holidays, and weather. She finds that throughout the academic year there was an upward trend in absences, and that winter storms caused the absolute highest total absences (Vald, 2024). Limitations include the collection time of the data (before the 2020 pandemic) and the exclusion of school districts’ attendance goals. Notably, depending on how data was sorted, a skew occurred towards schools with a larger/smaller population accordingly.

Furthermore, in the paper “The Chronic(les) of Absenteeism Measurement: Unpacking the Many Measures of Attendance and Evidence for a Lower Chronic Absenteeism Threshold”, Wu, Wiland, and Staines find that chronic absenteeism is associated with lower academic achievement and reduced graduation likelihood, and that empirically derived risk thresholds cluster in the 3–7% range—supporting our decision to treat absenteeism rate as continuous rather than applying a fixed binary (Wu & Staines, 2026).

The above research focuses on understanding absenteeism through an individual or environment focused lens. However, it does not address why chronic absenteeism exists institutionally or the structural drivers of such absenteeism in a public school system in the city. Our project bridges this gap by testing school-level NYC prediction with temporal validation and interpretable structural features.

3. Dataset and Evaluation

We draw on two publicly available NYC DOE Open Data sources: the [Attendance Results Dataset](#), and the [Demographic Snapshot School Dataset](#). These are joined on DBN and year.

The merged dataset spans from 2013-2018 and consists of 7,607 observations and 50 features, the majority of which appear in paired form, with each measure recorded as both a raw count and a percentage of total enrollment. As raw counts are not directly comparable between schools of differing sizes, all subsequent analyses use percentage-based representations exclusively, including the target variable: chronic absenteeism rate. Two features, Borough and District, were excluded from the merged dataset due to missingness rates of 95% and 99%, respectively.

Data Splits: To further explore the relationship between chronic absenteeism and the selected features, we experimented with two distinct splitting methods.

- **Randomized Splitting:** Using an 80/20 ratio and `seed(123)`, we split the data into a training and testing set, respectively.
- **Temporal Splitting:** We trained on 2013–2016, tuned on 2016–2017, and tested on 2017–2018 (60/20/20 split).

We implemented multiple linear regression and polynomial regression on both splits and implemented more advanced methods only for the temporal split, which better reflects real-world conditions.

Table 1. Row count summary after pre-processing.

STAGE	ROW COUNT	DESCRIPTION
TOTAL	7607	TOTAL ROWS AFTER CLEANING
TRAIN	4540	TRAINING DATA
TUNE	1530	DEVELOPMENTAL YEAR
TEST	1537	FINAL EVALUATION YEAR

We evaluate model performance using MSE , $RMSE$, R^2 , and R^2_{adj} . MSE and $RMSE$ capture average prediction error, with $RMSE$ being more interpretable. R^2 and R^2_{adj} measure explained variance, with the latter penalizing model complexity to facilitate fair comparison.

4. Methods

4.1. Baseline Methods

4.1.1. MULTIPLE LINEAR REGRESSION

Our baseline fits an OLS regression of % Chronically Absent on four structural predictors: Economic Need Index (ENI), % Poverty, % English Language Learners (ELL), and % Students with Disabilities (SWD).

4.1.2. POLYNOMIAL REGRESSION

Given the complexity of the data, we decided to use polynomial regression to determine which order model could best predict the percentage of chronically absent students. We extend the baseline by replacing the linear ENI term with a polynomial degree of j ($j = 1, \dots, 10$), while keeping the other three predictors linear. ENI has by far the highest t -value (34.381), suggesting that it is the most statistically significant predictor among the four features.

4.2. Machine Learning Methods

4.2.1. RANDOM FOREST

We implemented random forest on our four structural predictors, using 5-fold cross-validation to determine the best hyperparameters for `mtry` (number of variables sampled, range 1-4) and minimum node size (range 2-20) via a regular grid search with 5 levels per parameter, selecting the combination that minimized *RMSE*. The number of trees was fixed at 500 to balance performance and stability with computational cost.

4.2.2. XGBOOST

XGBoost was implemented on the same four structural predictors using 5-fold cross-validation to tune three hyperparameters: number of trees (100–1000), tree depth (3–8), and learning rate (0.001–0.1, searched on a log scale). A regular grid with 4 levels per parameter was used, balancing coverage of the search space with computational cost given the larger grid induced by three hyperparameters. The optimal combination was selected by minimizing *RMSE*.

4.2.3. SHAP ANALYSIS

For both methods, we computed TreeSHAP values on the test set to examine both aggregate feature performance and individual observation contributions.

5. Experiments

5.1. Diagnostics

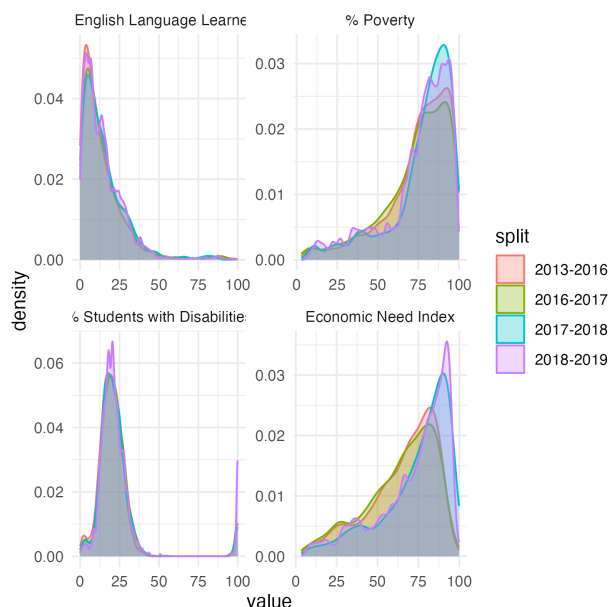
While choosing between a random or temporal split, we observed a discrepancy between the expected outcome and

the actual outcome of our random split’s train vs. test error.

Test error was significantly smaller than train error in the random split. We began examining our dataset to find potential causes of this. One possibility was data leakage. This was quickly ruled out by experimentation with a grouped random split, collecting schools so that they only appear once in either test or train, where the same phenomenon occurred.

We then adopted a new temporal split method with a test, tune, and train split. Across all models, tune error was much closer to train error, while test error was significantly larger than both train and tune. This led to the discovery that, in the 2017–2018 school year, a new data matching process that was implemented led to an increase in both ENI and % Poverty (NYSED, 2017). This was undocumented in our dataset, though noted in the 2017–18 to 2021–22 Demographic Snapshot dataset.

Figure 1. Density Plots of Predictors



The distributions of ENI and % Poverty shift notably after 2017–18, with ENI in particular showing a sharp mode increase from 72.2 in 2016–17 to 91.3 in 2017–18, consistent with the new data matching process described above. This rightward compression reduces predictor variance and likely limits model discrimination among high-need schools, as observations that would previously have been distinguishable by ENI are pushed toward the upper end of the distribution. The distributions of ELL and % Students with Disabilities remain comparatively stable across years.

KS tests confirm that ENI and % Poverty differ significantly

between train and test ($D = 0.233, p \approx 0$) and tune and test ($D = 0.241, p \approx 0$), while train and tune are statistically indistinguishable for both predictors. ELL and % Students with Disabilities show no substantial distributional shift between any splits. This isolates the 2017–18 methodological change as affecting specifically the economic need predictors, consistent with the ceiling compression observed in Figure 1.

These results offer a plausible explanation for the anomalous behavior observed under the random split, and for the divergence between tune and test error under the temporal split.

5.2. Multiple Linear Regression

Train yields an R^2 of 0.676 whilst tune and test R^2 stabilize at 0.50, indicating that the selected predictors explain approximately half the variance in chronic absenteeism. Though the random/grouped split data are not reported, they produced a lower test MSE than train MSE.

Table 2. RMSE values for MLR.

TRAIN RMSE	TUNE RMSE	TEST RMSE
10.608	10.572	10.95

Table 3. R^2 values for MLR.

	TRAIN	TUNE	TEST
R^2	0.676	0.507	0.506
R^2_{adj}	0.676	0.505	0.505

Figure 2. RMSE Plot by Polynomial Degree

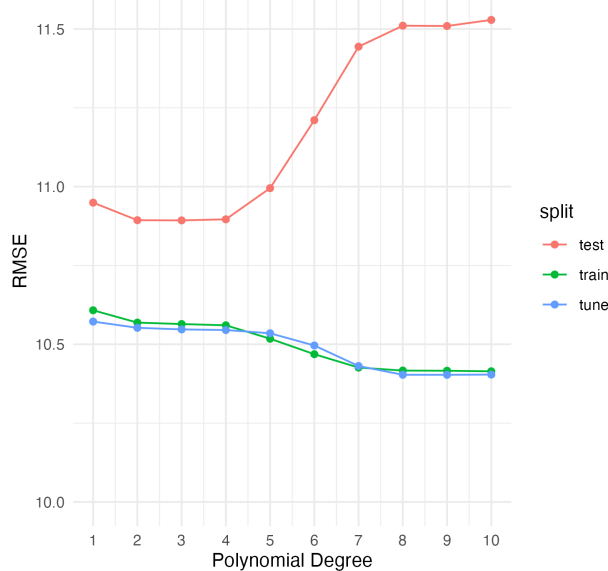
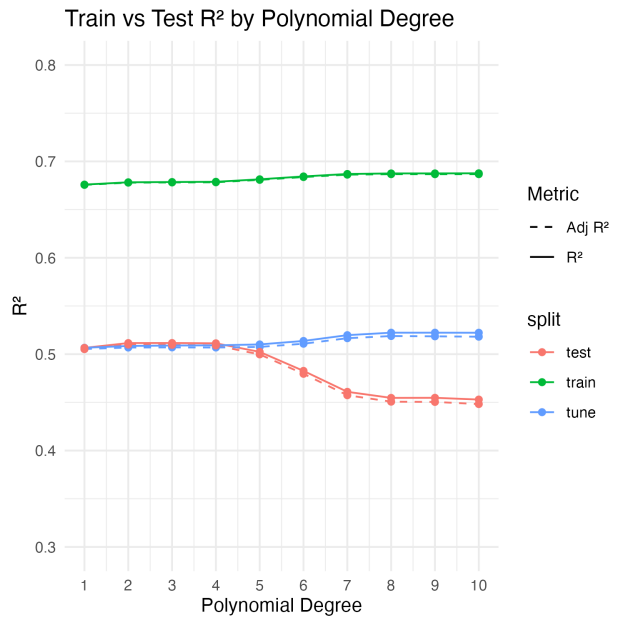


Figure 3. Train vs. Test: R^2 and R^2_{adj}



Train and tune RMSE remain close throughout, both decreasing monotonically with polynomial degree. Test RMSE sits substantially higher across all degrees and rises sharply after degree 5, consistent with overfitting compounded by the distributional shift identified in the diagnostics section. Although tune error is minimized at degree 9, we select degrees 3–4 as optimal, where the gap between all three splits is smallest and model complexity remains justified before test error begins its sharp increase.

5.3. Random Forest

The best hyperparameters for RF were found to be $mtry = 3$ and $min.n = 11$. Sampling three of four features at each split approaches the full feature set. The relatively large minimum node size of 11—toward the upper end of the search range of 2–20—suggests that cross-validation pushed toward greater regularization to improve generalization, yet even this was insufficient to prevent overfitting

Table 4. RMSE values for RF.

TRAIN RMSE	TUNE RMSE	TEST RMSE
5.769	10.026	11.386

Table 5. R^2 values for RF.

	TRAIN	TUNE	TEST
R^2	0.857	0.556	0.466
R^2_{adj}	0.857	0.555	0.465

RF exhibits the largest train-to-test discrepancy of all models

and clear signs of overfitting, with train RMSE of 5.769 rising to 11.386 on test and train R^2 falling from 0.857 to 0.466. While the model explains approximately 85% of variance in the training data, this drops to under 47% on the test set. Tune error sits in between (RMSE 10.026, R^2 0.556). This suggests that the model was already generalizing poorly prior to the 2017–18 distributional shift.

Figure 4. SHAP Breakdown for RF.

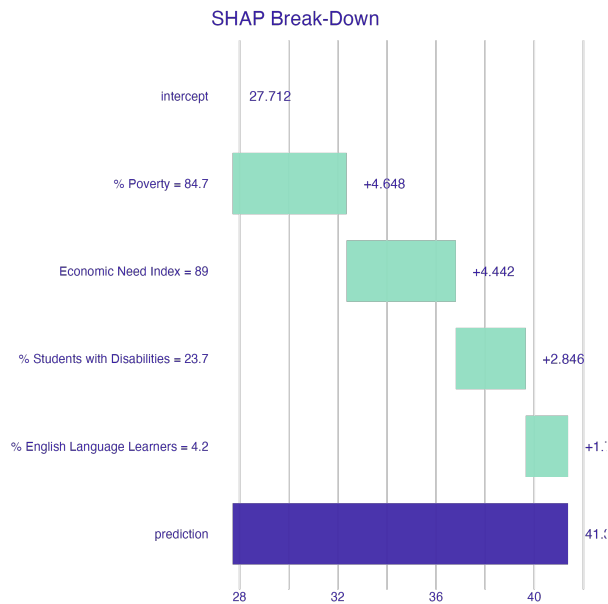
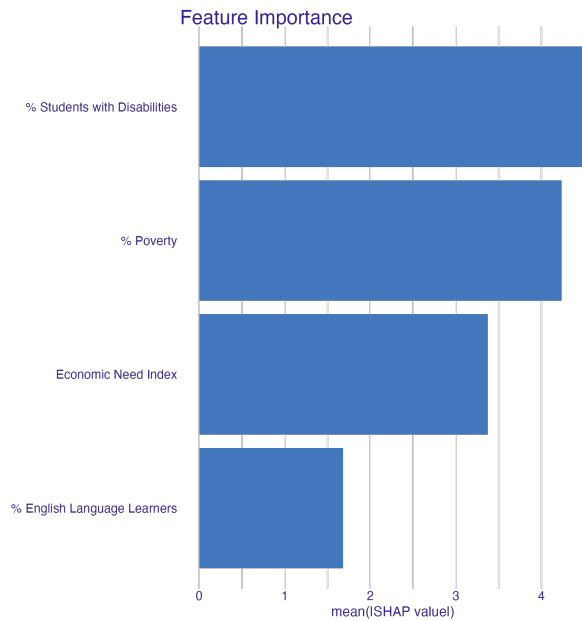


Figure 5. Feature Importance for RF.



% Students with Disabilities ranks as the most important

feature, with ENI falling to third. This is likely an artifact of RF distributing credit unpredictably across correlated features given the high mtry, rather than a substantive finding. The SHAP breakdown for a representative high-need observation (ENI = 89, % Poverty = 84.7, % SWD = 23.7, % ELL = 4.2) yields a predicted chronic absenteeism rate of approximately 41%, meaning RF predicts that 41% of students at this school are chronically absent—substantially higher than XGBoost’s prediction of 15.3% for the identical observation.

The divergence is largely attributable to RF’s substantially higher baseline prediction of 27.7% compared to XGBoost’s 2.6%, reflecting RF’s tendency to memorize the training distribution rather than learn generalizable structure.

5.4. XGBoost

The best hyperparameters for XGBoost were found to be *trees* = 100 and *tree_depth* = 4, and *learn_rate* = 0.1. Few trees and a high learning rate suggests that the model converged quickly, whilst the shallow depth suggests that each individual tree is capturing relatively simple relationships between the predictors. This is consistent with the limited complexity of a four-predictor problem, where three of the four features are correlated by construction—ENI is itself computed from % Poverty and ELL, meaning the model is learning from partially redundant signals rather than four independent sources of information. Additional model capacity, such as deeper trees or more boosting rounds, would therefore be unlikely to improve generalization and may in fact hurt it given the distributional shift in ENI and % Poverty identified in the diagnostics.

Table 6. RMSE values for XGBoost.

TRAIN RMSE	TUNE RMSE	TEST RMSE
9.280	9.697	10.736

Table 7. R^2 values for XGBoost.

	TRAIN	TUNE	TEST
R^2	0.630	0.585	0.526
R^2_{adj}	0.630	0.584	0.524

XGBoost achieves the most stable results across splits, with train, tune, and test RMSE of 9.280, 9.697, and 10.736 respectively (the smallest train-to-test gap of any model tested). The model explains between 52.6% and 63.0% of variance in chronic absenteeism across splits, with R^2 declining modestly from train to test.

Figure 6. SHAP Breakdown for XGBoost.

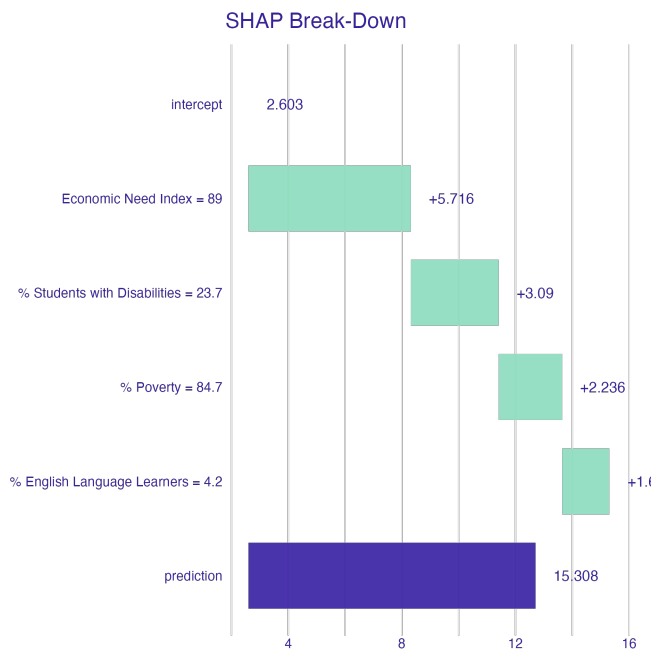
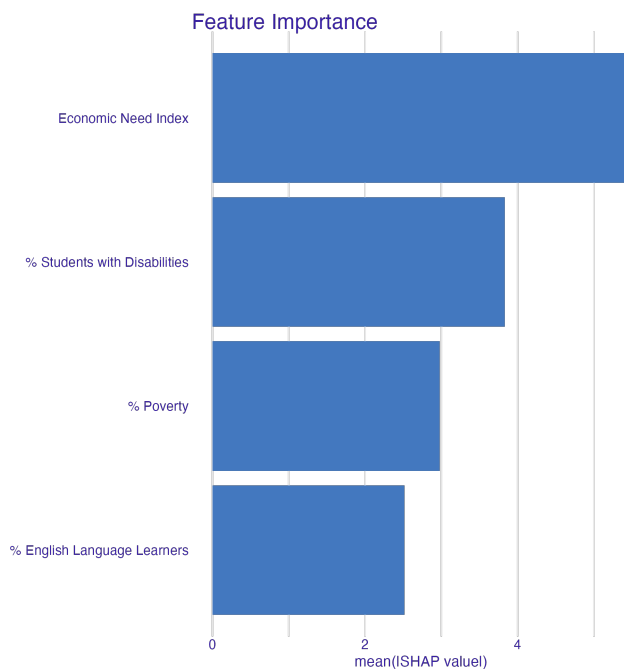


Figure 7. Feature Importance for XGBoost.



In the feature importance plot, ENI has the highest mean SHAP value, consistent with its role as an aggregate of the other economic predictors. Notably, % Students with Disabilities ranks second, ahead of % Poverty and % ELL, suggesting that it captures genuinely independent signal rather than serving as a redundant predictor. Students with disabilities may face greater barriers to attendance due to medical appointments, health-related absences, or other

disability-associated factors that are structurally distinct from economic need and not reflected in ENI, % Poverty, or % ELL.

For the same representative high-need observation as RF, XGBoost yields a baseline of 2.6% with ENI contributing +5.7, % Poverty +4.6, % SWD +2.8, and % ELL +1.6 percentage points, for a final prediction of 15.3%. Feature contributions are of comparable magnitude to those produced by RF, suggesting the two models agree on the direction and relative influence of each predictor. However, they disagree on feature ranking, which likely reflects RF's unstable credit allocation among correlated features rather than a substantive difference in the learned relationships.

6. Conclusion

Our models successfully identify the direction and magnitude of chronic absenteeism from structural variables, with XGBoost proving most reliable, achieving the smallest train-to-test RMSE gap across all methods while maintaining stable R^2 across splits. Across both models, ENI emerged as the dominant predictor by SHAP magnitude, though its susceptibility to administrative redefinition limits its reliability as a stable structural signal. Of the remaining predictors, % Students with Disabilities proved the most independently influential, which is notable given that ENI is itself computed from % Poverty and % ELL, meaning SWD captures genuinely distinct structural barriers to attendance not reducible to economic need.

After conducting error analysis by capping training ENI values at 95 to better match the test distribution, XGBoost's RMSE was reduced by 0.52 (shown in Figure 8.) and optimal tree depth dropped to 3, indicating that the distributional mismatch was a meaningful source of test error. RF's RMSE also improved under this adjustment, confirming the issue affected both models, though XGBoost remained more reliable overall.

This was somewhat surprising: cross-validation had pushed RF toward greater regularization via a larger minimum node size, which we expected to improve generalization. However, RF's high mtry meant it was still sampling nearly the full correlated feature set at every split, likely limiting the benefit of that regularization and leaving it more vulnerable to the distributional shift.

Future work should address model sensitivity to administrative changes in predictor definitions, for instance by incorporating a year $\times$ ENI interaction term to explicitly model temporal instability. A further direction is to extend the framework by incorporating graduation outcomes from the NYC DOE High School Performance Directories (2013–2018), which would allow the model to trace how structural absenteeism drivers translate into longer-term student out-

comes.

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A. Appendix

Figure 8. Clipping Effect on RMSE & R^2 for RF and XGBoost

